

We developed an ocean Wave Energy Harvester (WEC) to harness ocean energy. There are essentially two ways in which you can harness energy from the wave action - to pump fluid or by rotating a shaft. We successfully rotated a shaft using wave energy in our model. It may be mentioned that only this initial part is the invention. Subsequent steps are all well known science and existing technology i.e hydraulic transmission, conversion etc. As an extension of this technology, we have designed a wave powered boat, but not yet prototyped. This is a remote controlled boat with a small payload of one ton and with an extraordinary cruising range. This has many applications for offshore surveillance with long endurance times. We intend using this in our Ocean cleanup effort in some time.

We have developed a silt trap that passively traps silt and this is removed by dedicated conveyors. This is a very important invention since silt is a major maintenance expense for drains, rivers, dam reservoirs or ports. Some basic solutions already exist for trapping silt, but obviously that is not good enough and the problem persists all around us. This is a low operating cost model and seeks to continuously control depths, rather than restoring depths with expensive dredging or desilting.

We have developed the latest generation of marine oil spill response equipment i.e integrated systems with containment and skimming functions incorporated. It may be very interesting to note that there is hardly any perimeter fencing on the waterside. Most waterside security systems are acoustic intrusion alarms. But the alarm helps only when you have a response team standby 24x7. We made a strengthened version of these barriers as a boat barrier for perimeter security along with floating gates. Waterfront facilities like shipyards and dam reservoirs would typically like these solutions.



Q What have been some of the most significant challenges you've encountered in your river cleanup projects? How have you overcome these challenges?

A City river clean-up is challenging. The rivers have shallow depths, wide monsoon flows, fast flows in monsoon and very narrow dry season flows, rocky river bed etc. Mechanical equipment from the river bank cannot work across the river width. You do not get floating cranes in these rivers. But we realise that we cannot change the river morphology and hence we adapt to the challenges each and every day. Vandalism and pilferage in remote locations is another issue and we have now successfully addressed this issue after undergoing some expensive learning.

Q Have you collaborated with local governments, communities, or other organisations? How have these partnerships influenced or accelerated your cleanup efforts?

A Since we are working in lakes and rivers, invariably we work with the government. It is interesting to see in

close up, how the Government functions. Government has a very big canvas, a big pocket and consumes practically anything that is produced.

Working with the Government, subjects you to longer cashflow cycles - with some remarkable exceptions - since the Govt is very process oriented and this process is loaded with checks and balances. Anyone in business knows that liquidity is very important. Now 'time' as a resource is equally important. So we had many cash-stressed times and struggled to keep the RnD momentum going. But we persevered, and yet generated a portfolio of very practical and scalable technologies. It was certainly a very rough ride, but we have crossed that bridge.

Government of Goa has willingly supported AlphaMERS' proposal to clean up all plastics from all the rivers and streams along the entire coast of Goa. This will be the largest such project in the world, funded by corporations, using cost effective indigenous (AlphaMERS) technology, and active support of the Government in disposing the collected trash. M/s Investindia has come forward to carry out an impact assessment of the entire project. This ambitious project is planned to be completed in 2 years time. This will ensure no plastics drain into the ocean from the Goan coast and also less of plastics that will be carried by littoral currents and deposited on the state's beaches.

Our own expertise lies in arresting and bringing trash and plastics to the riverbank at a small cost. We prefer to focus on developing better technologies for such interventions, where others do not have solutions. We do not get into the transport of waste or processing of the waste, which by itself is a vast subject and there are others with the necessary expertise. We communicate better with technology. We leave the community interaction to others with social sector expertise. **d**